

Questions ☐

1. Predict the output

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    while ('1' < '2')
        cout << "In while loop" << endl;
}
```

2. Predict the output

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int t = 10;
    while (t /= 2) {
        cout << "Hello" << endl;
    }
}
```

3. Predict the output

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    for (int x = 1; x * x <= 10; x++)
        cout << "In for loop" << endl;
}
```

4. Predict the output

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int x = 10, y = 0 ;
    while ( x >= y ) {
        x-- ;
        y++ ;
        cout << x << " " << y << endl ;
    }
}
```

5. WAP to print the sum of all the even digits of a given number.

Sample Input : 4556
Output: 10

6. WAP to print the sum of a given number and its reverse.

Sample Input : 12
Sample Output : 33 [12+21]

7. Print the factorials of first 'n' numbers

Sample Input : 10

Output :

```
1
2
6
24
120
720
5040
40320
362880
3628800
```

8. Print first 'n' fibonacci numbers.

Sample Input : 10

Output :

```
1 1 2 3 5 8 13 21 34 55
```

9. Write a program to print out all Armstrong numbers between 1 and 500. If the sum of cubes of each digit of the number is equal to the number itself, then the number is called an Armstrong number. For example, $153 = (1 * 1 * 1) + (5 * 5 * 5) + (3 * 3 * 3)$

Output :

```
1
153
370
371
407
```